# Bits and bytes

## IoT · Level 2 · Lesson 3



# Today we will

- Count in binary with place values, from 0 to 255.
- Explain why more bits give finer readings (10 bits give 1,024 levels).
- Estimate how much data a reading, a message and a photo take.

# Counting with switches

- A bit is 1 or 0
- Each place is worth double
- 8 bits = 1 byte = 0 to 255
- More bits: finer readings

# Binary dot cards

1. Warm-up (5 min): Compare an Arabic and an English SMS of the same length.
2. Teach (8 min): Bits, place values, and reading 101 and 110.
3. Cards (12 min): Groups make five dot cards and race to show numbers.
4. Apply (10 min): Pupils write their age in binary and check a partner's.
5. Wrap-up (5 min): Why can't one byte hold 300?

**Make one set of cards to show the class. The dots must be clear from the back of the room: 16 dots fit in a 4 × 4 grid.**

# Let’s discuss

1. Where do your pupils meet bits and bytes in daily life?
2. Why do data bundles count megabytes, not messages?
3. What would a sensor need more bits for?

# Exit ticket

- Write 13 in binary, and say why 8 bits stop at 255.